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1. Data overview

- Life expectancy (`lex.csv`): 194 countries × 301 yearly columns (1800 onward). `geo` (3-letter code), `name`, then one column per year. Values are years of life expectancy at birth.
- Employment rate (`aged_15plus_employment_rate_percent (1).csv`): 179 countries × 36 yearly columns (1991 onward). Same `geo`/`name` structure. Values are the % of the population aged 15+ that is employed.

5. Interpretation

Across countries in 2020, life expectancy and the 15+ employment rate are essentially uncorrelated at the cross-country level. A few takeaways:

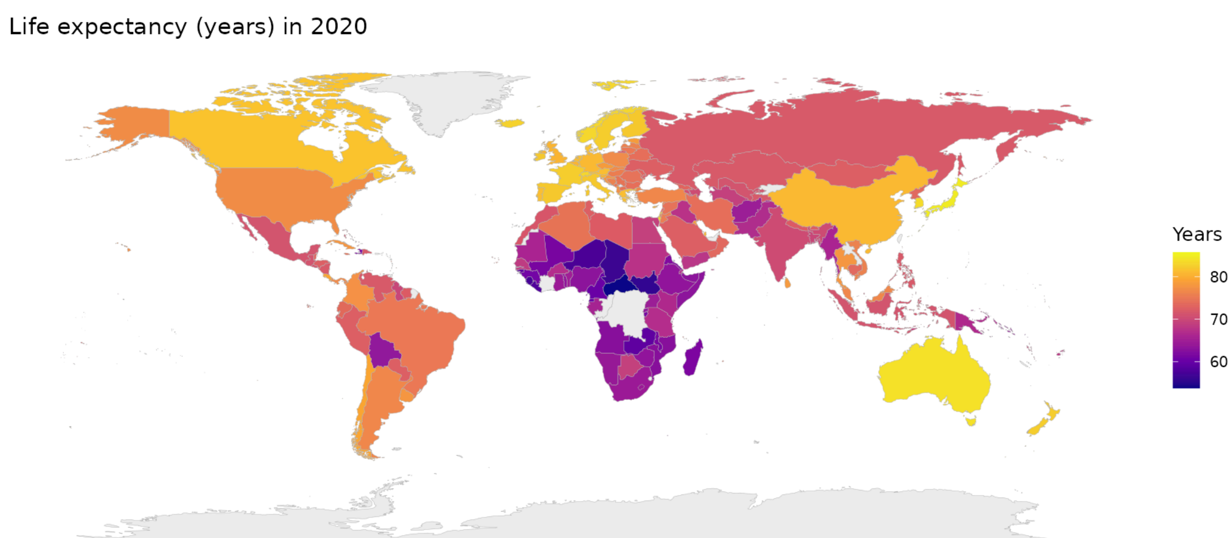
- The slight negative slope is not meaningful ($r \approx 0$). Knowing a country's employment rate tells you almost nothing about its life expectancy, and vice versa.
- The cloud is wide and U-shaped if anything: many rich, long-lived countries (Western Europe, Japan) have moderate-to-low employment rates because of longer schooling, earlier retirement, and generous welfare systems, while many lower-income countries (parts of sub-Saharan Africa, South/Southeast Asia)

have very high employment rates driven by subsistence and informal work — yet shorter life expectancy.

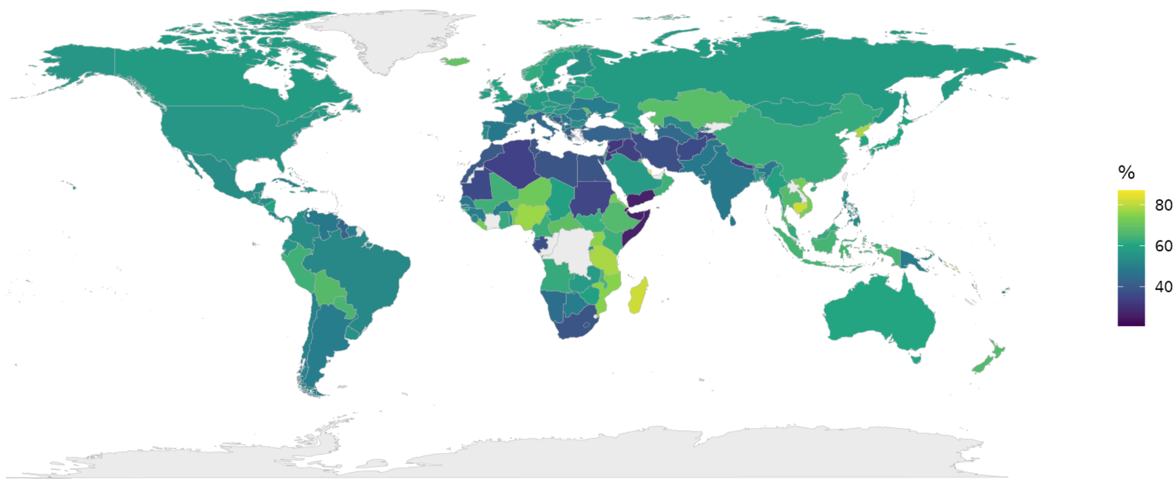
- High-income, high-employment outliers also exist (e.g., Gulf states, Singapore, Switzerland), which further muddies any simple linear story.
- The real driver of life expectancy is the level and quality of income, healthcare, nutrition, and education — not the share of adults who happen to be employed. Employment rate captures *labor-market structure*, which is shaped by demographics, welfare regimes, and informality, not by health outcomes.

Bottom line: in 2020, employment rate is not a useful predictor of life expectancy on its own. For a meaningful health-outcomes model you'd want GDP per capita, healthcare spending, education, and sanitation — with employment rate as, at most, a contextual control rather than a primary explanatory variable.

The country-level scatter shows essentially no linear relationship: Pearson r = −0.016.



Employment rate (age 15+) in 2020



Reflection

I actually had to go through multiple prompts to get Julius to give me the graph and the maps in a proper format. It gave me nothing and then invalid URLs, and then finally gave me downloadable images.

Overall, It was pretty easy to use. I expected more iteration to be need or for it to ask for guidance. Some tools I have used ask follow up questions and maybe that is just the way I am prompting and I did not include anything like that here.

I like the thinking being displayed because I could see julius going through the document and that helped me be more confident in the findings.